

Experimental verification of the egress and ingress latency correction in PTP clocks

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Abstract—Precise time synchronization using the Precision Time Protocol (PTP, IEEE 1588-2008) is based on time-stamping of data packets on egress and ingress. The accuracy of the time-stamps is highly affected by the ingress and egress latencies of the PTP node. For an accurate synchronization it is therefore crucial to know the characteristics of these latencies. In our paper we present a measurement setup that enables the direct measurement of both the ingress and egress latencies of a PTP clock. In contrast to our earlier work, the principle utilizes standard PTP peer delay mechanism messages. It can therefore be applied to any PTP clock that supports the peer delay mechanism. It can thus verify whether the egress and ingress latencies of commercially available devices are correctly compensated. In our contribution we give a detailed description of the measurement setup. We demonstrate the feasibility of the system by presenting results for several devices with different time-stamping hardware.

Index Terms—IEEE 1588, hardware propagation delay, PTP, synchronization, time-stamping

I. INTRODUCTION AND BACKGROUND

The necessity to have a common time base throughout a distributed system which is interconnected via an unreliable network was addressed by designing synchronization protocols [1]. Synchronizing the clocks of all devices of the system is achieved by continuously adjusting the local clocks to the time of a common time master [2]. One such synchronization protocol is the Precision Time Protocol (PTP) defined by IEEE 1588 [3]. By using PTP, it is possible to achieve accuracies in the sub-microsecond and even sub-nanosecond range, if certain requirements are fulfilled. One important prerequisite for accurate synchronization is the capability of the hardware to deliver precise time-stamps on PTP network packets. Any kind of delay, jitter, or asymmetry introduced to the time-stamp values has a high impact on the synchronization accuracy [4]. Since the whole synchronization concept of PTP is based on the assumption that the network propagation delay is symmetrical (which in reality is not), it is important to know any kind of asymmetries and to compensate them.

As required by clause 7.3.4.2 of [3], *the time-stamp shall be the time at which the event message time-stamp point passes the reference plane marking the boundary between the PTP node and the network*. Nonetheless, depending on the hardware and software capabilities of a PTP node, time-stamps can be generated in different locations on the path of a PTP message (see Figure 1). In general, the closer to the reference plane the time-stamps are generated, the better

the synchronization accuracy becomes [5]. So, we defined the egress delay to be *the delay from the actual time-stamping point to the wire* and the ingress delay to be *the delay from the wire to the actual time-stamping point*, like in [6] and [7].

The data sheets of several hardware chips that we have consulted ([6], [7], and [8]) present propagation delays for different PHYs from different vendors. Unfortunately, it is not always clear under which conditions these delays were measured or simulated. Some of the data sheets present a maximum and a minimum for the delay, implying the existence of a jitter, which in some cases ranges in the hundreds of nanoseconds. Under these circumstances we decided to look not only for possibilities on how to measure the egress and ingress delays of our own hardware, but also for possibilities on how to verify the propagation delay for commercial off-the-shelf (COTS) products. [9] presents a measurement setup on how to measure the propagation delays and their variation depending on different 100Base-TX Ethernet PHY chips. In order to find out the start of the Ethernet frame, they had to limit the test setup to 100 Mb/s half-duplex connections with the auto negotiation disabled. Further, [10] presents vendor specific propagation delays, and finds out that the transmit and receive delays of the PHYs vary by approximately a factor of 3. None of these methods we looked so far could be used to measure delays for COTS devices. Since the egress and ingress delays highly depend on cable type, interface and connection speed, we argue that a black box approach to reliably measure them, even for off-the-shelf devices, is needed.

In our earlier work, we presented a measurement setup that allowed the direct measurement of egress and ingress delays of a PTP clock [4]. The setup is based on an oscilloscope capable

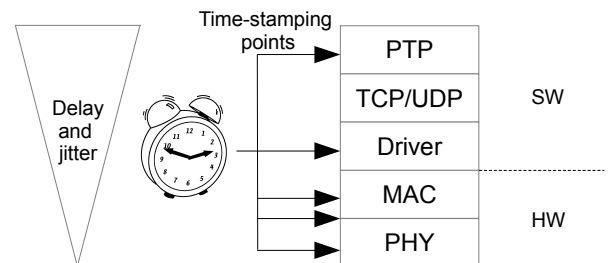


Fig. 1: Time-stamping points in a PTP node [4]

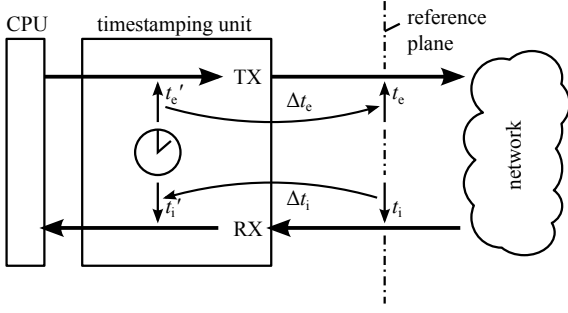


Fig. 2: Ingress delay Δt_i and egress delay Δt_e .

of decoding Ethernet traffic. Using such an oscilloscope, time-stamps for PTP event messages can be taken on the wire. By comparing these oscilloscope measurement with the hardware time-stamps generated by the device under test (DUT), the ingress and egress delays of the device can be obtained. The measurement setup however has several shortcomings. First of all, the measurements are limited to 10Base-T and 100Base-TX Ethernet, since the oscilloscope is only capable of decoding packets on such connections. Since a loop-back cable is used in the measurement setup, the system is limited to full-duplex connections. The most severe problem however is caused by the fact that a customized software must be running on the DUT. This requirement can easily be met with prototype devices during development, but inhibits the setup from being used for the verification of COTS devices from third-party vendors.

In this work we present an extended measurement setup for the measurement of the egress and ingress delays of PTP clocks. We focus on a system that can be used for the verification of COTS devices, on which a customized software for the measurement cannot be run. The paper is organized as follows: section II presents the measurement principles. In section III we discuss the measurement setup and focus on some design issues and limitations. Finally, section IV gives measurement results for several ordinary clock devices.

II. MEASUREMENT PRINCIPLES

Figure 2 depicts a typical setup for a system that supports hardware time-stamping. The device comprises a time-stamping unit in the network path. This unit is typically integrated into the Ethernet controller (MAC), into the Ethernet PHY, or in an FPGA that is located between MAC and PHY. The time-stamping unit detects egress and ingress PTP event messages. If an event message passes the time-stamping unit, a time-stamp is generated by recording the current time on the internal PTP clock. Time-stamps which are generated on egress messages are labeled t'_e in Figure 2, whereas the time-stamp t'_i is generated on ingress messages.

Since time-stamping of the messages is done at some point off the reference plane, t'_e and t'_i are affected by egress and ingress latencies, Δt_e and Δt_i . To compensate for these delays and to obtain the reference plane time-stamps t_e and t_i the

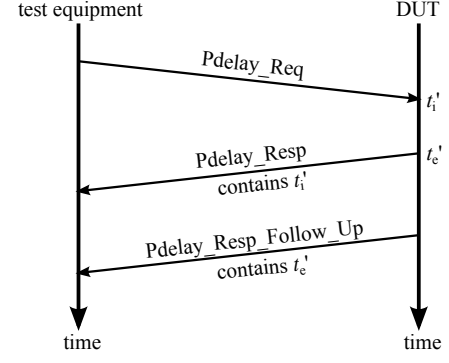


Fig. 3: Obtaining time-stamps from the DUT using peer delay messages. The ingress time-stamp t'_i is generated on the arrival of the Pdelay_Req message. It is transferred to the test equipment in the Pdelay_Resp. When the Pdelay_Resp passes the time-stamping unit of the DUT, the egress time-stamp t'_e is taken and made available to the test equipment using the Pdelay_Resp_Follow_Up message.

corrections:

$$\begin{aligned} t_e &= t'_e + \Delta t_e \quad \text{and} \\ t_i &= t'_i - \Delta t_i \end{aligned} \quad (1)$$

must be applied (see 7.3.4.2 in IEEE 1588-2008 [3]).

To be able to apply the corrections of (1), the delays Δt_e and Δt_i must be known. In [4] we presented a setup that measured the time-stamps t_e and t_i of egress and ingress Sync messages directly at the reference plane with an oscilloscope. Together with t'_e and t'_i obtained by the DUT, the ingress and egress latencies could be calculated,

$$\begin{aligned} \Delta t_e &= t_e - t'_e, \\ \Delta t_i &= t'_i - t_i. \end{aligned} \quad (2)$$

However, this setup cannot be applied for the verification of COTS devices, since a customized software must be executed on the DUT to obtain t'_e and t'_i of the Sync packets.

To allow the setup to be used for COTS devices, we have to utilize data packets whose time-stamps are revealed by the DUT within the standard PTP protocol. For the egress time-stamp, the Sync message fulfills this criterion. The egress time-stamp of a Sync message is either included in the message itself (one-step) or subsequently sent using a Follow_Up message (two-step mode). To obtain an ingress time-stamp from a DUT, the Delay_Req message would be a good candidate. Its ingress time-stamp is returned to the network in a Delay_Resp message. Unfortunately, choosing Sync/Follow_Up and Delay_Req/Delay_Resp to get time-stamping data from the DUT limits the application of the setup to master-capable clocks. Slave-only ordinary clocks ([3], clause 9.2.2) will never enter the MASTER state. They will therefore not send Sync messages and will not respond to Delay_Req messages.

Since the power profile [11] is one of our main applications, all of the DUTs in the scope of this paper supported the peer delay mechanism (P2P). Within this mechanism, we can also

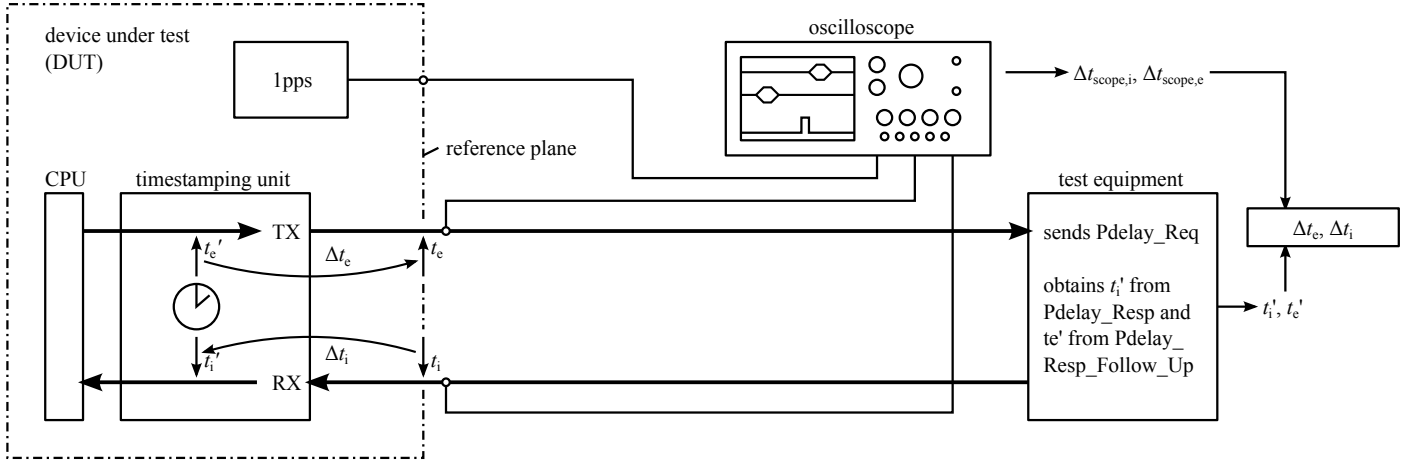


Fig. 4: Measurement setup

find messages that potentially reveal the time-stamps taken by the DUT. In a P2P two-step clock that operates according to [3], 11.4.3, option (c) 8, Pdelay_Req messages sent to the DUT are time-stamped on arrival (see Figure 3). A corresponding time-stamp value t'_i is subsequently published in a Pdelay_Resp message. This t'_i may already be corrected for the ingress delay Δt_i by the DUT, but in the considerations below we will just ignore this possibility and treat timestamps from the DUT as if they were uncompensated or at least wrongly compensated. Doing so will allow us to use the measurement setup for the verification of the delay compensation of the DUT.

The Pdelay_Resp message mentioned above is time-stamped on egress, yielding a t'_e that is also made available by the Pdelay_Resp_Follow_Up message (also shown in Figure 3). Since both slave and master clocks respond to the messages of the peer delay mechanism, using them allows the measurement principles to be applied to both slave and master clocks. However, it cannot be applied to clocks supporting the end-to-end mechanism only. Furthermore, one-step clocks do not provide t'_i and t'_e directly, but add the turnaround time $t'_e - t'_i$ to the correction field of the Pdelay_Resp message. Therefore, the principle is only usable for clocks that operate in two-step mode for Pdelay_Resp messages. For two-step mode, the standard allows two more options in 11.4.2 of [3]: Option (c) 8 directly yields the two time-stamps of interests, whereas option (c) 7 just returns the turnaround time in the correction field. Consequently, the measurement principle is further limited to clocks that operate according to 11.4.3 option (c) 8 of [3] and publish the time-stamp values directly. However, the devices that we tested fulfilled both conditions. They operated as two-step clocks for peer delay messages. Furthermore, clause 5.3 of the power profile [11] advises devices to use option (c) 8 of clause 11.4.3 of the IEEE 1588 standard [3]. Therefore, many devices in our main field of application will also fulfill this condition.

III. MEASUREMENT SETUP

A. Overview

With t'_e and t'_i obtained from the DUT, we still need to measure the times at which the data packets actually pass the reference plane, t_e and t_i . In [4] we used a simple loopback cable and connected an oscilloscope to the data pair via a differential probe. Since the loopback cable was very short, t_e and t_i were equal and could be obtained from a single delay measurement. Using the measurement principles proposed above, a simple loopback cable can no longer be used. Instead, the DUT must be connected to a test system that operates as a PTP ordinary clock. The test system will establish PTP synchronization with the DUT. Most important, it will send Pdelay_Req messages periodically, and collect time-stamping data from the Pdelay_Resp and Pdelay_Resp_Follow_Up messages (Figure 3) the DUT replies. To be able to perform timing measurements for both incoming and outgoing messages with the oscilloscope, we consequently need two differential probes connected to both pairs of the Ethernet line.

Figure 4 shows the device under test (DUT) on the left hand side. It comprises a CPU that runs a PTP stack, a hardware time-stamping unit that includes the PTP clock, and a 1pps output to generate a one-pulse-per-second (1pps) signal. The Ethernet jack of the DUT marks the reference plane between the device and the network. The time-stamps t_e and t_i shall be generated when a PTP event message passes this reference plane. The DUT is connected to the test equipment via an Ethernet cable. This cable was modified to allow connecting the differential probes of an oscilloscope to both data pairs as close to the Ethernet jack as possible. The distance between the Ethernet jack and the probes is only a few centimeters, we will therefore neglect the small error caused by this short cable length. The differential probes are connected to channel 1 and 2 of the oscilloscope. Furthermore, the 1pps signal from the DUT is connected to the third channel of the scope. The oscilloscope that is used in our measurements is a Tektronix DPO4054B. It is equipped with the DPO4ENET option that allows decoding Ethernet traffic.

The oscilloscope is configured to trigger on the rising edges of the 1pps signal, which marks the start of a second. Furthermore, the oscilloscope decodes the Ethernet data on both channel 1 and channel 2. Figure 5 shows an example of an oscilloscope measurement. The measurement shows that both a Pdelay_Req and Pdelay_Resp packet have been captured. In addition also the Pdelay_Resp_Follow_Up message can be seen in the screen. The trigger point in the center of the screen marks the beginning of a second. It establishes a relationship between the oscilloscope traces and the internal clock of the DUT.

The oscilloscope measures the delays between the edge of the 1pps signal and the start of the Ethernet preamble of the egress and ingress event messages. These measurement results had to be adjusted by the length of both the preamble and the start of frame (SOF) delimiter to obtain the delays between the 1pps signal and the time-stamping points as defined by IEEE 1588 (see [3], 7.3.4.1). On 100 Mb/s Ethernet transmitting the preamble and the SOF delimiter takes 640 ns. Hence, 640 ns must be added to both delays, yielding the corrected values $\Delta t_{\text{scope},i}$ and $\Delta t_{\text{scope},e}$.

Both these delays and the content of the data packets are automatically obtained using the oscilloscope's event table feature. By issuing the commands SAV:EVENT:B1 and SAV:EVENT:B2 to the scope, the event tables can be stored to memory. They can then be transferred to the controlling computer using the FILES:READF command.

From the delays $\Delta t_{\text{scope},i}$ and $\Delta t_{\text{scope},e}$ we will later calculate the time-stamps t_e and t_i , that mark the time at which the event messages pass the reference plane. But first we will discuss some details of the measurement setup.

B. Message rate

An oscilloscope measurement can only cover a limited window in time. In our measurement setup we operate the oscilloscope at a sample rate of 1.25 GS/s, which corresponds

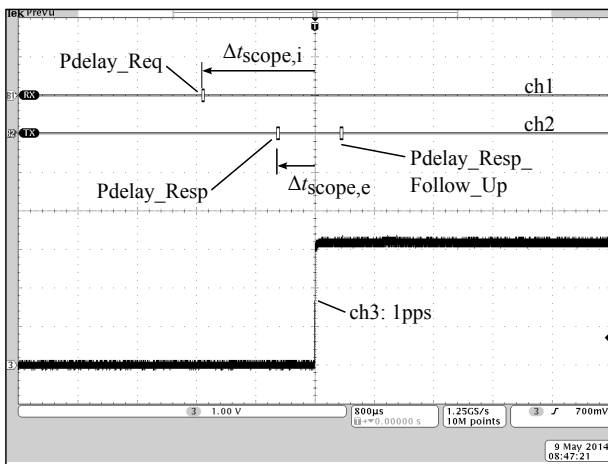


Fig. 5: Screenshot of an oscilloscope measurement. The results of this sample are $\Delta t_{\text{scope},i} = -1.511287$ ms and $\Delta t_{\text{scope},e} = -0.495490$ ms.

to an 800 ps resolution. The scope is configured to acquire 10^7 samples in a single measurement, thus covering a time interval of

$$\tau = 800 \text{ ps} \cdot 10^7 = 8 \text{ ms.} \quad (3)$$

Consequently, Pdelay_Req and Pdelay_Resp messages must appear on the Ethernet wire within a time frame of 8 milliseconds around the 1pps pulse. Otherwise, the oscilloscope cannot determine the time delay between the begin of a second and the appearance of the data packets on the wire.

Since the Pdelay_Req messages are generated by the test equipment, we could modify the software to send a Pdelay_Req close to the begin of a second. Since the DUT is required to respond with a Pdelay_Resp *as soon as possible after the receipt of the associated Pdelay_Req message* (from [3], clause 11.4.2), chances are good that both the response and the request lie within the same oscilloscope measurement.

However, this was not necessary for our measurements. Instead, we increased the message rate for Pdelay_Req messages, i.e. we set portDS.logMinPdelayReqInterval on the test equipment to -4. On our system, this value led to a message rate of approximately 15 messages per second, $R = 15 \text{ s}^{-1}$. Hence, the probability for a Pdelay_Req message to be detected by the oscilloscope is

$$R \cdot \tau = 15 \text{ s}^{-1} \cdot 8 \text{ ms} = 12 \% \quad (4)$$

Since Pdelay_Resp messages occur at the same message rate, the same probability calculation applies for these messages.

C. Duration of a measurement run

Unfortunately, the decoding of the Ethernet data on the oscilloscope takes quite long (up to 10 seconds until the packet data is displayed on the screen). Furthermore, the programming interface of the scope does not provide any means to query the decoding state. The program steering the oscilloscope measurement can therefore not determine if the decoding of the data is finished. Instead, fixed delays had to be introduced in the program. Taking a single oscilloscope measurement and transferring the data from the oscilloscope to the computer therefore takes approximately 15 seconds. Thus, about 240 oscilloscope measurements can be recorded per hour. However, only 12% of these measurements actually include data packets that can be used for egress or ingress delay calculations. This fact reduces the number of useable measurements per hour to $240 \cdot 12 \% = 28.8$. A long measurement duration is required to produce a sufficient number of samples. But since the setup is fully automated, the measurement can be run unattended over night.

D. Overrun of the sequenceId

When finally processing the measurement data, the oscilloscope measurements ($\Delta t_{\text{scope},i}$ and $\Delta t_{\text{scope},e}$) for a Pdelay_Req and Pdelay_Resp packet must be matched with the time-stamp data (t'_i and t'_e) from the corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up packet, respectively. In a first attempt the sequenceId was used to match Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up packets.

TABLE I: Measurement results for the egress and ingress delays of a prototype device for 100Base-TX Ethernet.

Row	DUT	Direction	Samples	min. [ns]	max. [ns]	mean [ns]	median [ns]
1	DUT 1: DP83640 PHY (from [4])	egress Δt_e	473	-19.8	-16.6	-18.6	-18.8
		ingress Δt_i	473	232.6	235.8	234.6	234.8
2	DUT 1: DP83640 PHY	egress Δt_e	482	-21	-14	-18.0	-17
		ingress Δt_i	482	225	239	231.7	232
3	DUT 2: P1010/VSC8572	egress Δt_e	640	-111	-50	-83.2	-97.0
		ingress Δt_i	633	949	1041	995.8	996.8

However, at a message rate of about 16 messages per second, the 16 bit wide sequenceId will overflow every 70 minutes. Since measurement durations of several hours are required to obtain a sufficient number of samples, an additional criterion is required to match the data packets. Our data processing software therefore not only compares the sequenceId, but in addition uses the seconds field of the originTimestamp, the requestReceiptTimestamp, and the responseOriginTimestamp (see clause 11.4.3 in [3]) to detect corresponding packets.

E. Data processing

In our measurement setup (Figure 4) the device labeled *test equipment* (TE) collects the t'_i and t'_e time-stamps from the DUT. The TE outputs a list of PTP messages that includes their message type (see 13.3.2.2 of [3]), sequenceId, and the time-stamp that the message contains (requestReceiptTimestamp or responseOriginTimestamp). From the oscilloscope, we get the delays $\Delta t_{scope,i}$ and $\Delta t_{scope,e}$ and the content of the corresponding PTP messages, which allows us to extract message type and sequenceId as well. In a first step we match Pdelay_Req messages captured by the oscilloscope with Pdelay_Resp messages received by the TE. From this matching we get pairs of $\Delta t_{scope,i}$ and t'_i . $\Delta t_{scope,i}$ is the time delay between a 1pps pulse and the time at which the data packet was detected at the reference plane (see Figure 5). Since the 1pps pulse was generated at the beginning of a second and the data packet was time-stamped at $t'_i = T'_{i,sec} + T'_{i,nsec} \cdot 10^{-9}$ (where $T'_{i,sec}$ is the seconds and $T'_{i,nsec}$ is the nanoseconds part of the time-stamp), which is a point in time that is within 8 ms around the 1pps pulse, we conclude that the absolute time at which the pulse was generated is

$$t_{pps} = \begin{cases} T'_{i,sec} & \text{for } T'_{i,nsec} < 500\,000\,000 \\ T'_{i,sec} + 1 & \text{for } T'_{i,nsec} \geq 500\,000\,000 \end{cases} \quad (5)$$

refer also to [4]. Adding the delay measurement results of the oscilloscope yields the time-stamp of the data packet on the cable,

$$t_i = t_{pps} + \Delta t_{scope,i}. \quad (6)$$

In the same way we match Pdelay_Resp messages decoded by the oscilloscope and the Pdelay_Resp_Follow_Up messages from the TE. From the packet time-stamp t'_e we obtain the absolute time of the 1pps pulse analogously to equation (5). Adding the delay measurement from the oscilloscope yields

$$t_e = t_{pps} + \Delta t_{scope,e}. \quad (7)$$

Using equation (2) the ingress and egress delays Δt_i and Δt_e can now be calculated.

F. 1pps output delay

The measurement method is based on the 1pps output signal of the DUT. This 1pps however is subject to internal delays and quantization errors. When the edge of the 1pps signal is delayed by Δt_{tr} , both $\Delta t_{scope,i}$ and $\Delta t_{scope,e}$ (see Figure 5) are decreased by the same value Δt_{tr} . Therefore, also the time-stamps t_e and t_i are decreased by Δt_{tr} , see (6) and (7). Finally, equation (2) shows that the presence of a 1pps delay Δt_{tr} decreases the egress delay Δt_e and increases the ingress delay Δt_i by the same absolute value — a fact that we should keep in mind when evaluating the measurement results. For some applications, e.g. obtaining compensation values for a prototype device, it may also be convenient to define $\Delta t_{tr} = 0$, see [4] for further details.

IV. MEASUREMENT RESULTS

A. Egress and ingress delays of a prototype device

During the development phase of a new PTP-capable product correction values for the ingress and egress delays must be obtained. In [4] we showed measurement results for a device based on the DP83640 Ethernet PHY [12], [6]. Those results are listed for comparison in row 1 of Table I. Using the extended measurement setup presented above we decided to repeat those measurements. We expected similar results as obtained in [4], in order to prove the feasibility of the new setup.

We connected the DUT to the measurement equipment as shown in Figure 4. The DUT operated as a master clock, whereas the test equipment synchronized to the DUT. The test equipment sent Pdelay_Req messages and recorded the time-stamp values from the DUT's responses. Delay measurements were obtained from the oscilloscope as described above.

The total duration of this measurement was 18 hours, 13 minutes. Within this time the oscilloscope took 4703 measurements, a measurement every 14 seconds. 482 Pdelay_Req and 482 Pdelay_Resp messages were captured by the oscilloscope. Thus, 10.2 % of the shots taken by the scope contained Pdelay_Req and Pdelay_Resp messages. This number is less than we expected, see equation (4). It indicates that the generation of Pdelay_Req messages by the test equipment was not uniformly distributed in time.

From the time-stamps and the delays measured by the oscilloscope, the ingress and egress delays Δt_i and Δt_e were calculated. The results are listed in Table I, row 2. The results show good agreement between the mean and median values of the results of both measurement setups. The results

TABLE II: Measurement results for a COTS device that already compensates egress and ingress delays for 100Base-TX Ethernet.

Row	DUT	Direction		Samples	min. [ns]	max. [ns]	mean [ns]	median [ns]
1	DUT 3: COTS grandmaster	egress	Δt_e	67	-26	-10	-19.3	-19
		ingress	Δt_i	78	-22	-9	-15.1	-15

of the earlier measurement principle [4] exhibit a slightly smaller jitter, which may be attributed to the fact that a simple loopback cable was used in this setup. In contrast, the current setup utilizes a full Ethernet connection with a peer device and therefore reflects real-world conditions better.

Row 3 of Table I lists results for another prototype device based on a Freescale P1010 CPU and a Vitesse VSC8572 PHY. Although this Ethernet PHY is capable of PTP time-stamping, this feature was not used on the DUT. Instead, The time-stamping of PTP packets was done in the P1010's Ethernet controller. In contrast to DUT 1, which operated as a master clock, DUT 2 was a slave-only clock. The measurement thus proves the applicability of the measurement setup to PTP slave devices. Furthermore the results show larger delays and more jitter since time-stamping is done at a point farther off the reference plane.

B. Verification of a third party device

In this measurement the goal was to verify the delay compensation of a commercially available PTP grandmaster clock. Since a COTS device should already compensate correctly for its internal delays, the expected measurement result is 0 for both Δt_i and Δt_e .

The DUT was connected to the measurement setup as shown in Figure 4: The 1pps output of the DUT was connected to channel 3 of the oscilloscope, and the differential probes at channel 1 and 2 listened to the Ethernet traffic as close to the DUT's Ethernet jack as possible. The DUT operated as a PTP master, whereas the test equipment was a slave and synchronized to the DUT. Table II presents the measurement results. After the compensation of ingress and egress delays that is done internally by the DUT, negative delays of -15 and -19 nanoseconds remain, indicating that the device is overcompensating the delays.

V. CONCLUSIONS

One important requirement for a highly accurate PTP synchronization is the knowledge of the propagation delays through the hardware. This is the reason why a measurement setup was developed to find out the exact egress and ingress delays of a PTP node or to verify the fact that COTS devices correctly compensate the asymmetries introduced by the hardware.

The innovative measurement setup uses an oscilloscope that can decode and precisely time-stamp Ethernet frames. It was possible to use the presented measurement setup with COTS devices. We were able to calculate the egress and ingress delays by using the data obtained by the oscilloscope together with standard PTP messages, which were used to retrieve

time-stamping information from the DUT. This gave us the possibility to verify the delay compensation of third party products.

Unfortunately, the measurement setup depends on the characteristics of the oscilloscope. As stated in [4], similar drawbacks apply also to our extended setup. The used oscilloscope probe (DPO4ENET) supports only 10Base-T and 100Base-TX Ethernet, which makes a test with 1 Gb/s connections impossible. We reduced the oscilloscope's sample rate (at the cost of lower time-stamping resolution), and increased the synchronization rate of the DUT, in order to increase the probability of catching a PTP message of interest in the oscilloscope's measurement range. Even with these drawbacks, we were able to achieve our goal. The presented modifications enabled us not only to measure the egress and ingress delays of our own prototypes, but also verify the delay compensation of COTS devices.

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